Transition disks dynamics under non ideal MHD effects

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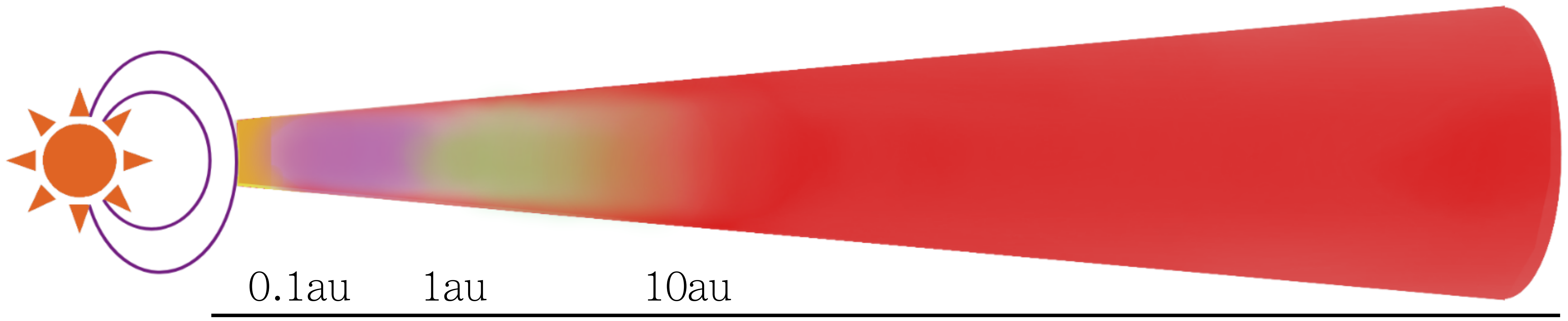
introduction

non ideal MHD effects

Protoplanetary disks (PPDs) are weakly ionised, $\chi \lesssim 10^{-10}$ - “frozen-in condition” no longer valid

Three species in plasma: neutrals, ions and electrons. Neutrals dominate $\rightarrow$ fluid velocity is $\mathbf{v}_n$ - for the rest we consider their relative drift



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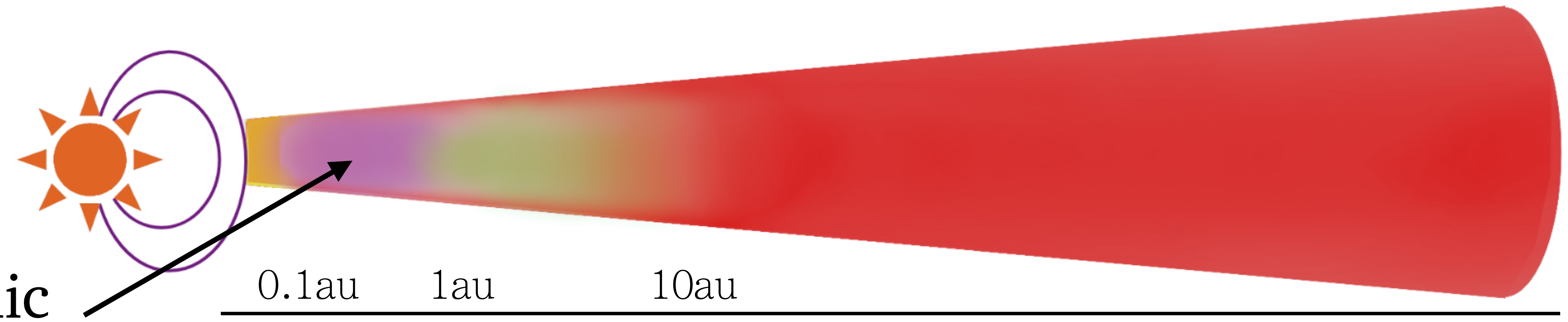

Ohmic resistivity ($e^- - n$ collisions)

current dissipation $\rightarrow$ dampening of magnetic field

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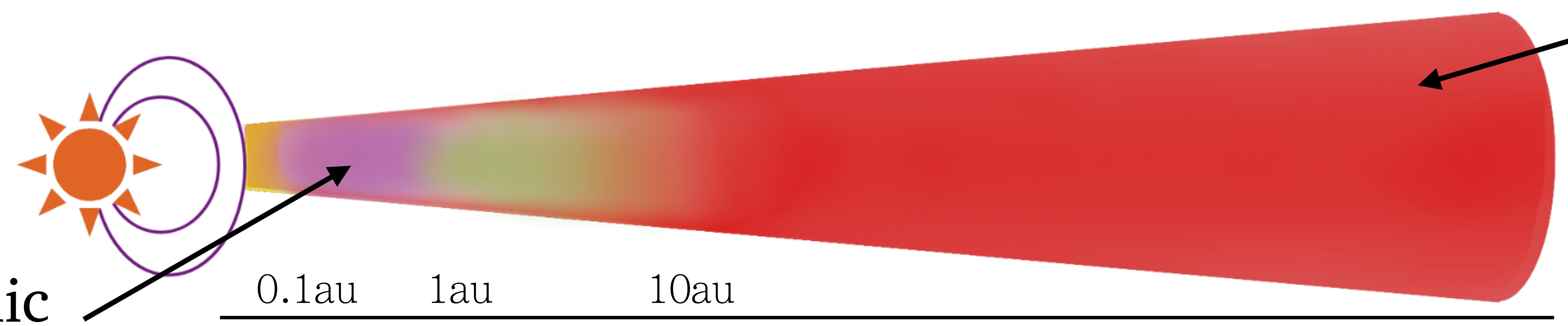

ambipolar diffusion ($i-n$ drift)

$n_n \gg n_i \rightarrow$
infrequent $i-n$ collisions $\rightarrow$
n move independently of magnetic field $\rightarrow$
weakened effect of m.f. to the flow

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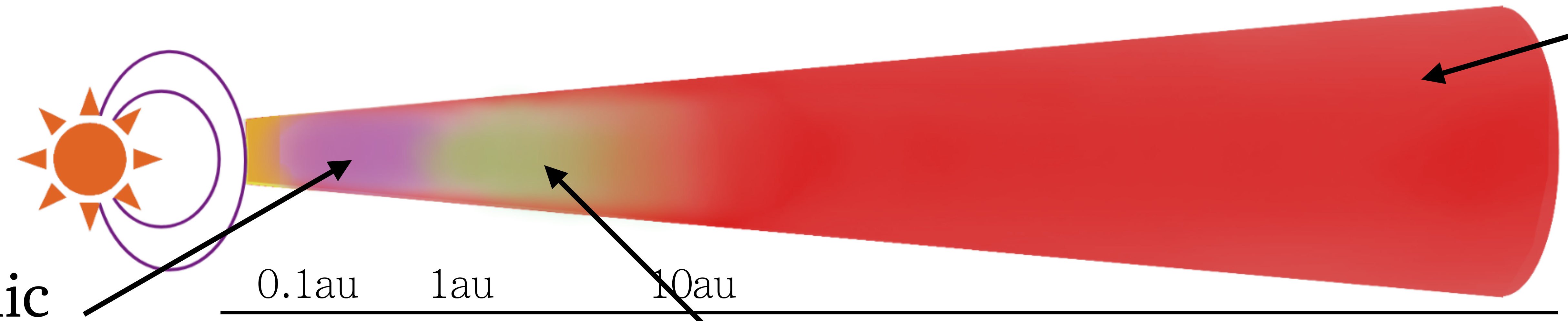

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Hall effect
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charge separation, drift between $e^- - i \rightarrow$
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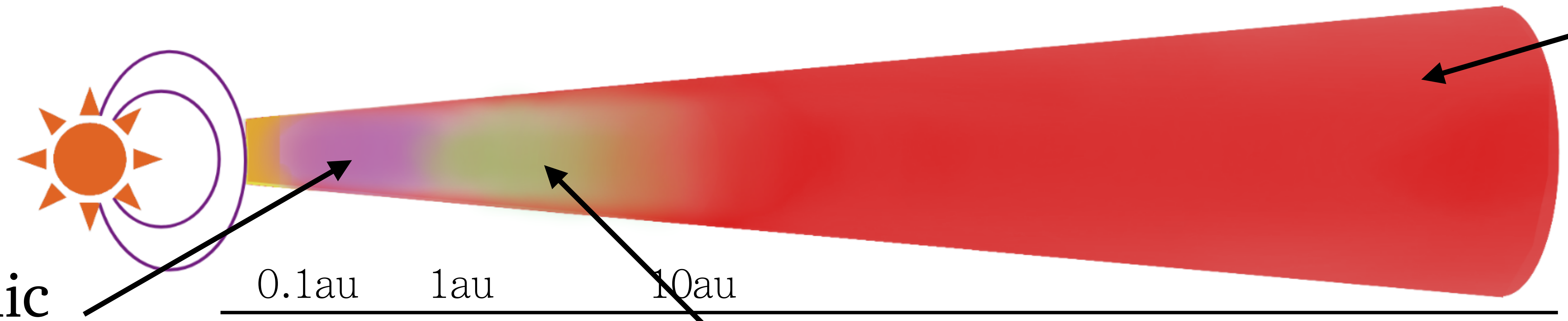

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$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) - \nabla \times \left[\eta_O (\nabla \times \mathbf{B}) + \eta_A (\nabla \times \mathbf{B})_\perp + \eta_H (\nabla \times \mathbf{B}) \times \hat{b} \right]$$

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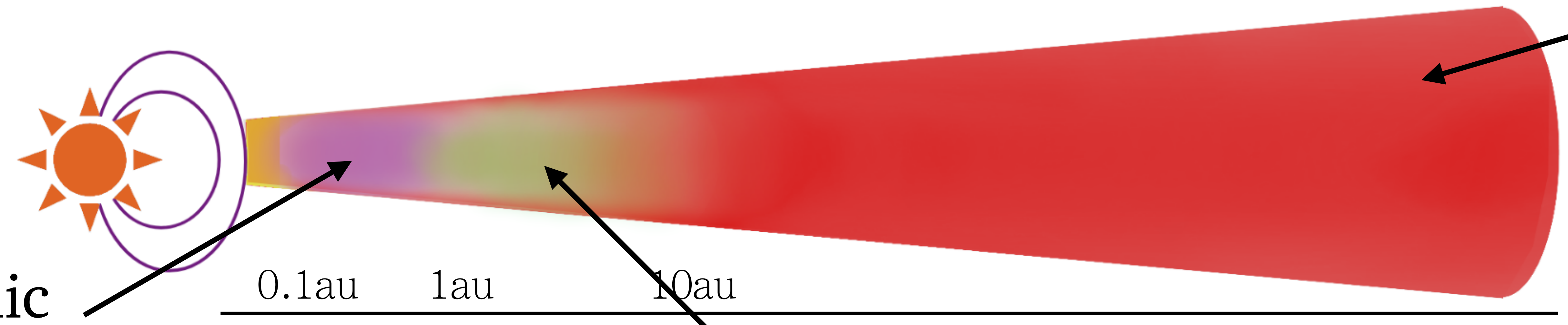

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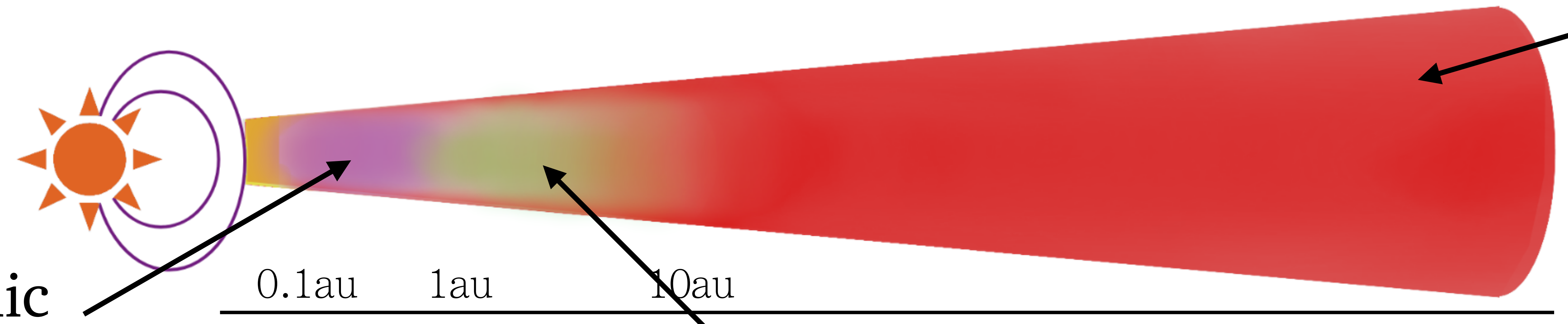

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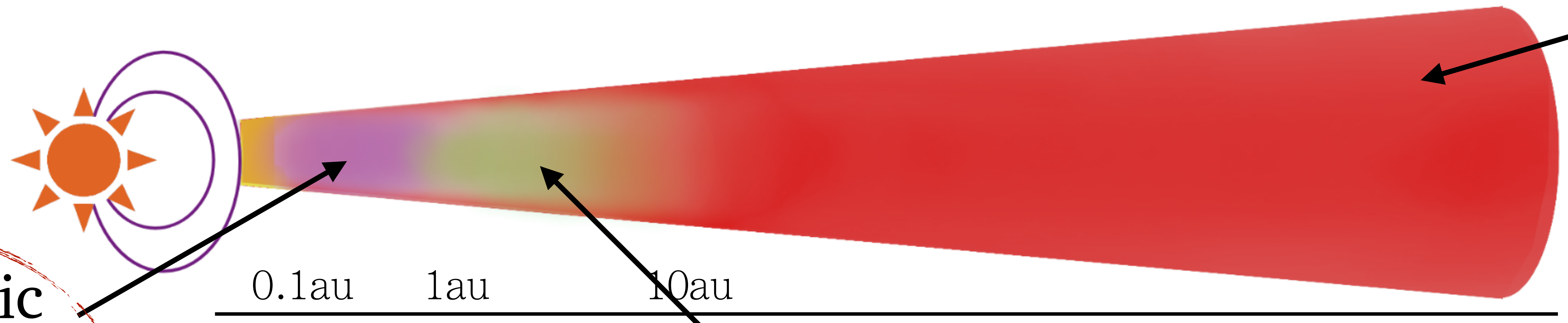

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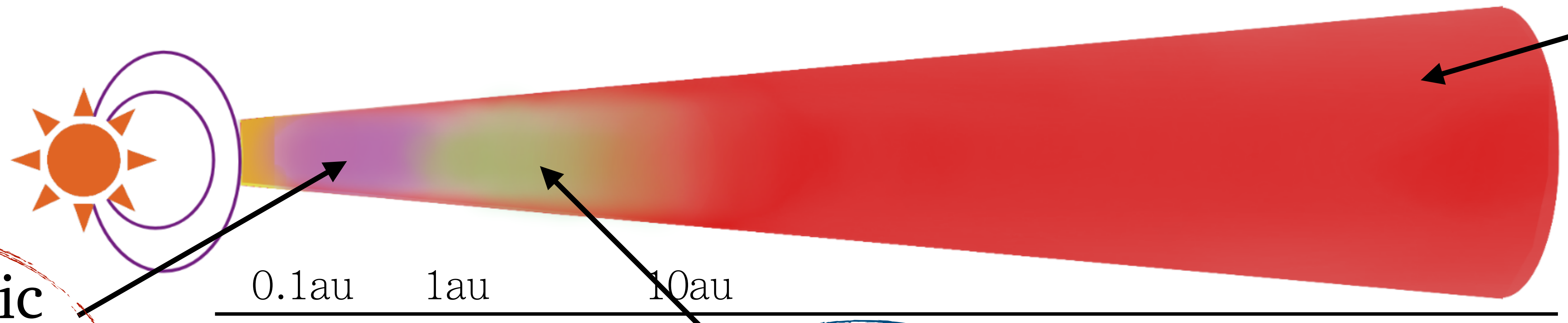

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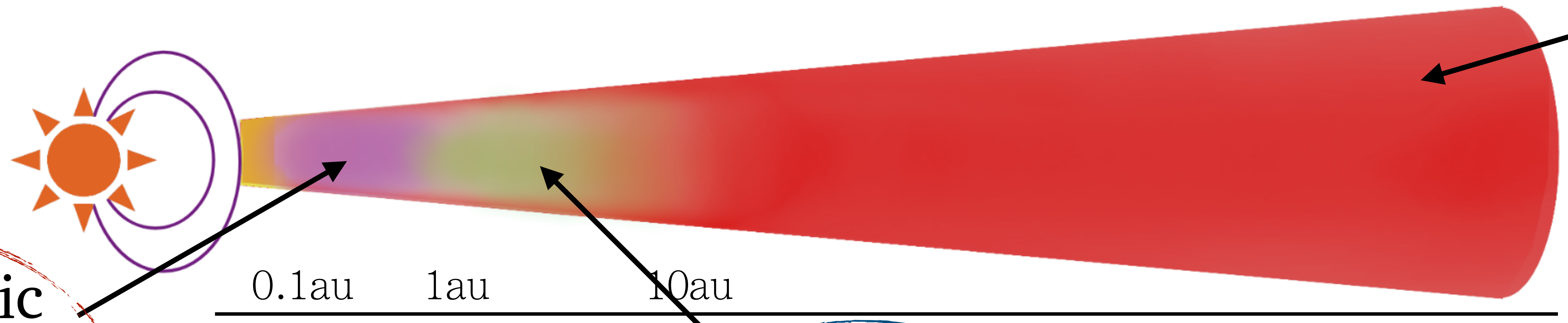

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HE dependent to polarity of $\mathbf{B}$:

- *Hall anti-aligned (h-aa)*:
  If $\mathbf{B} \downarrow\uparrow \hat{z}$, field lines tend to be straightened
- *Hall aligned (h-a)*:
  If $\mathbf{B} \uparrow\uparrow \hat{z}$, displacement of field lines is enhanced

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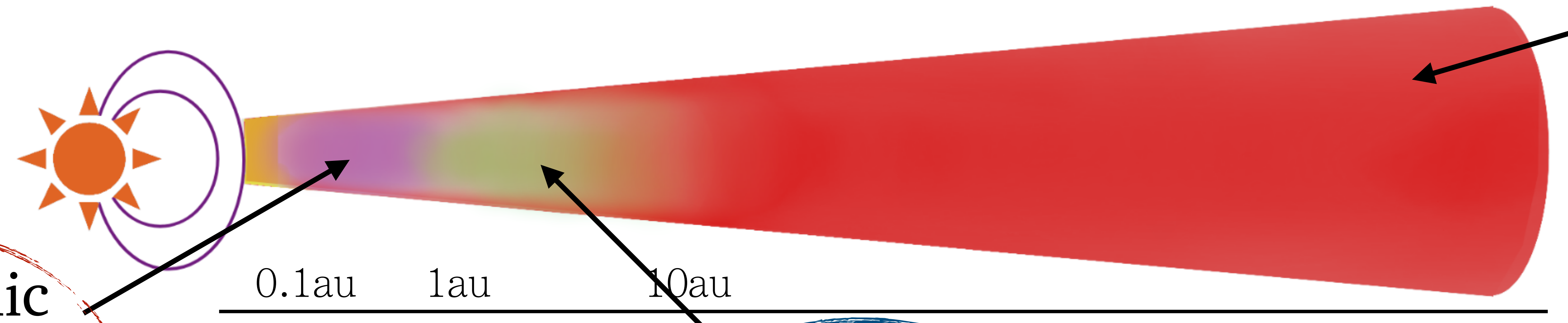

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- Numerous works about OR and AD:
  see eg. *Bai and Stone 2011,2013 , Gressel et al. 2015,2020, Lesur 2014, Hu et al. 2018, Suriano et al. 2018,2019, Wang et al. 2019, Cui & Bai 2021* etc.
- Less so about HE
  but see eg., *Bai 2017, Bethune et al. 2017, Krapp et al 2018, Lesur 2021*

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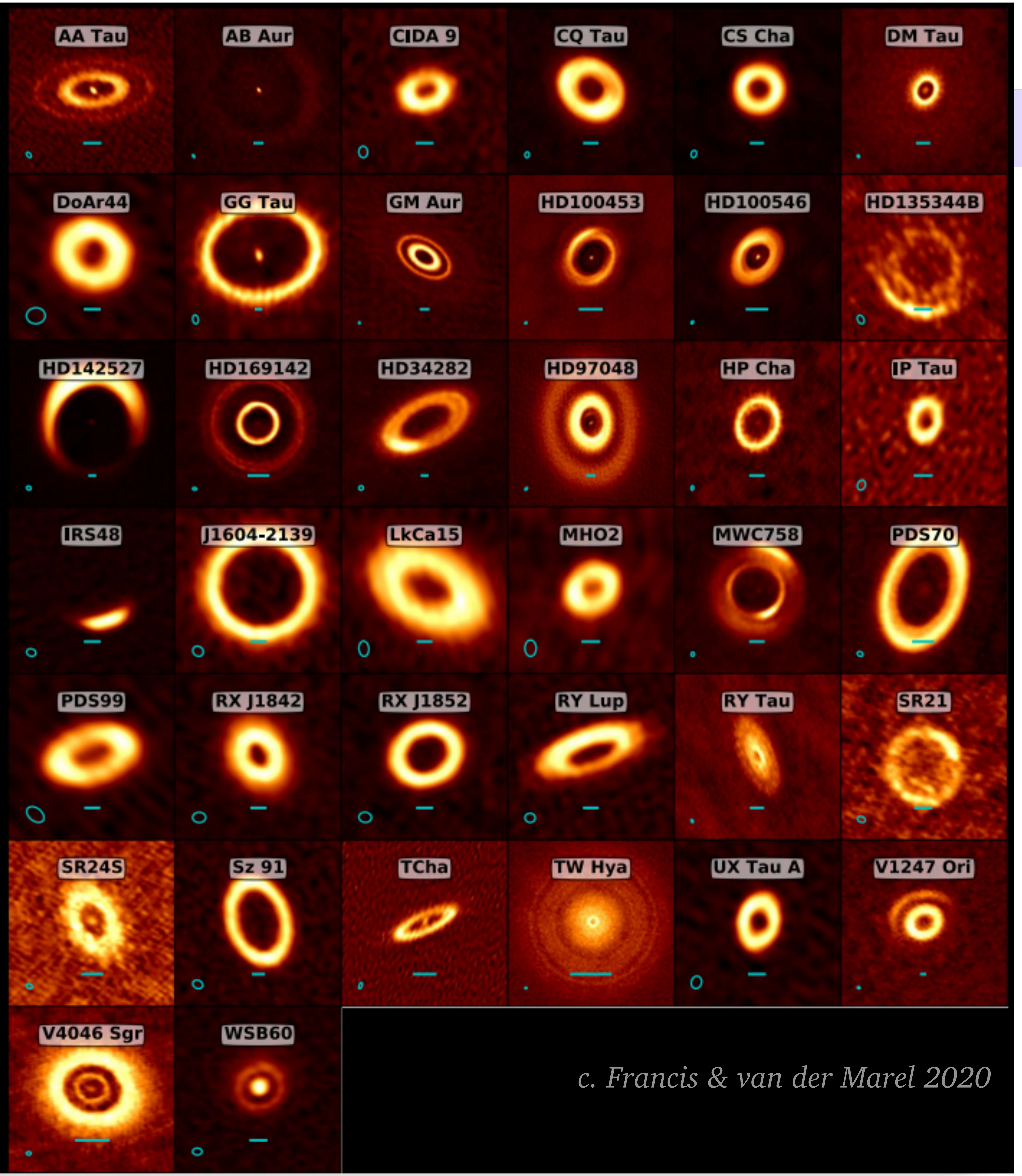
transition disks

- TDs are a class of disks characterised by inner dust and gas cavities
- Cavities could reflect a specific disk morphology (see eg. *Wang & Goodman 2017, Ercolano & Pascucci 2017, van der Marel 2023*)
- Very few works on TDs+magnetic fields, only one with global simulations (see *Martel & Lesur 2022*)



c. Francis & van der Marel 2020

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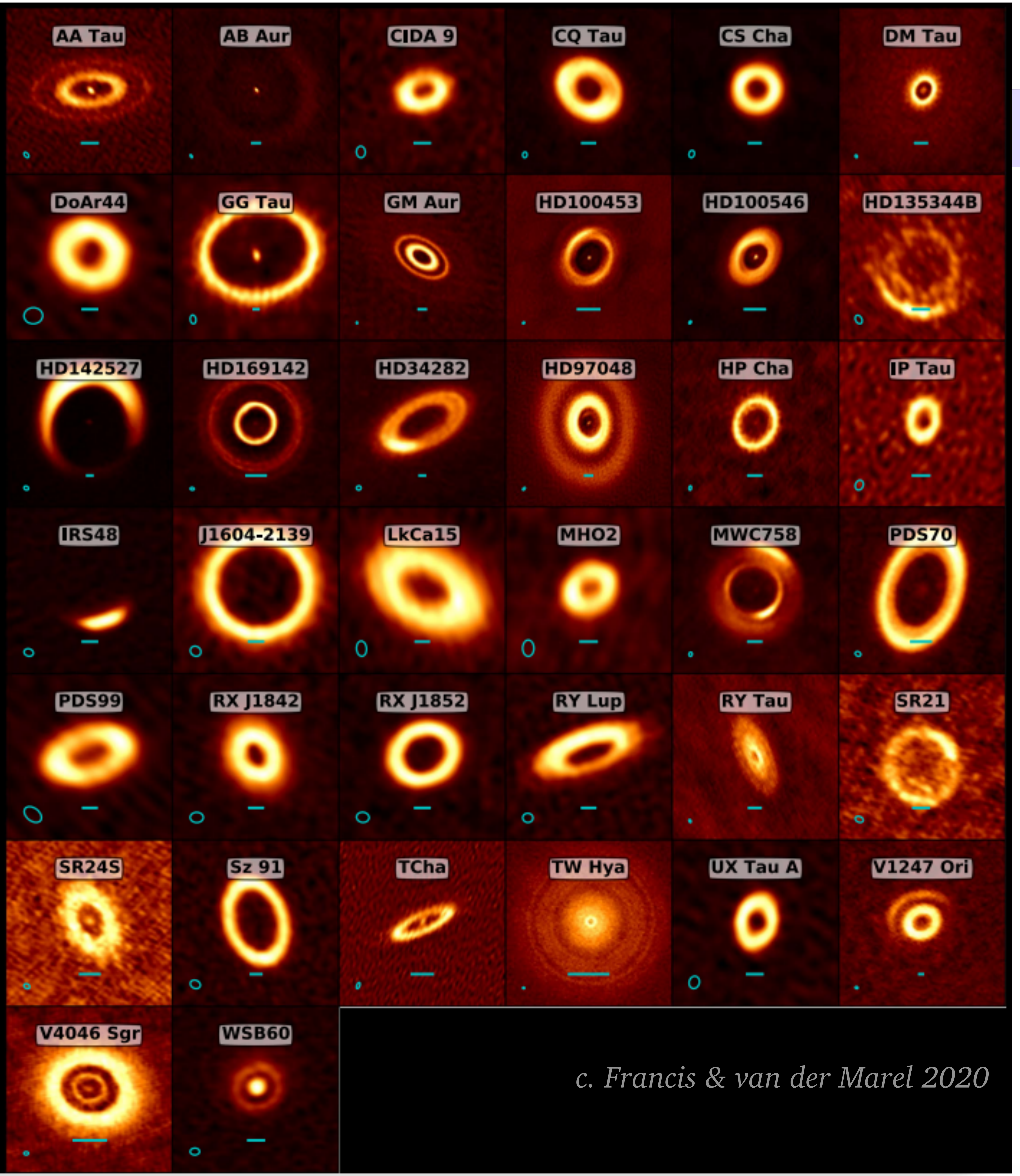
- 2.5D axisymmetric global simulations, focus on inner disk, 1-15 au
- 3 niMHD effects w. diffusivities calculated on the fly per grid cell
- Radiative transfer
- Assume a cavity

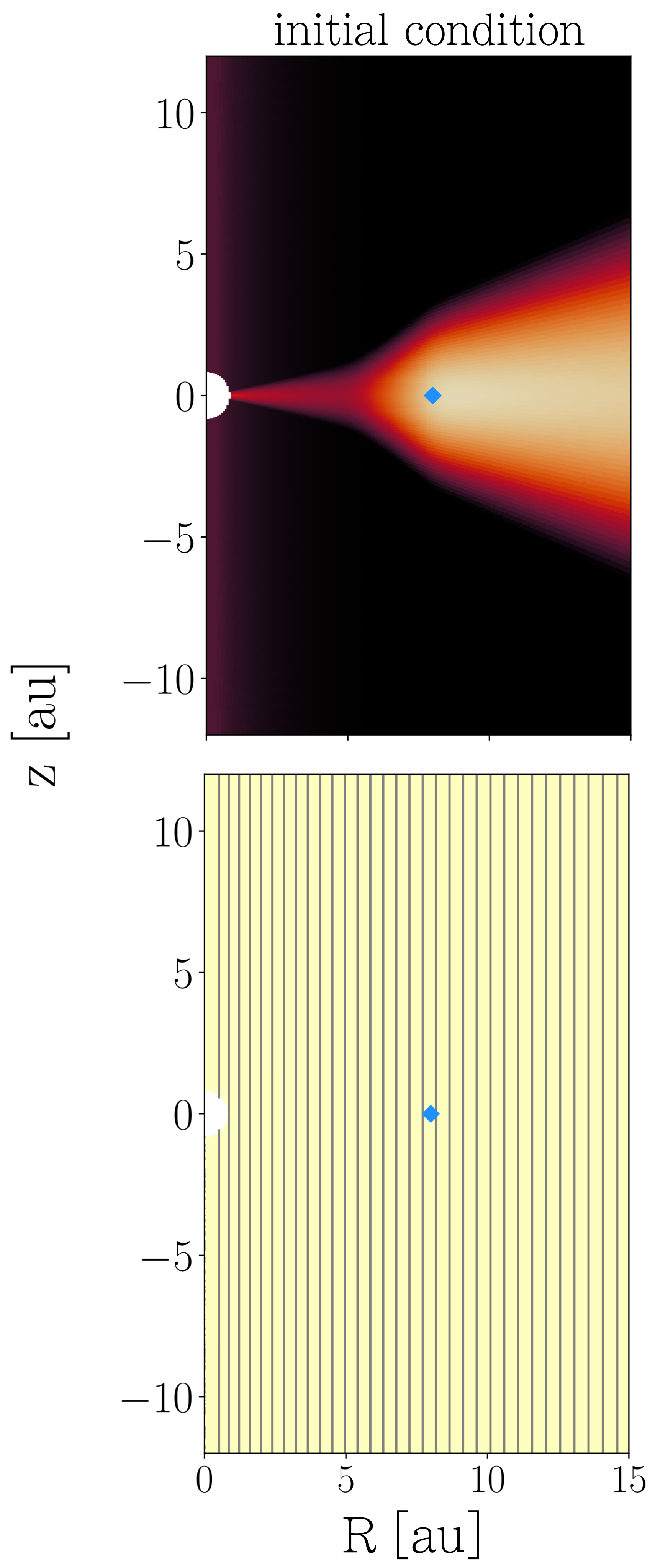


c. Francis & van der Marel 2020

results

general observations

h-aa : Hall anti-aligned, $\mathbf{B} \downarrow\uparrow \hat{z}$
h-f : only OR and AD, Hall free
h-a : Hall aligned, $\mathbf{B} \uparrow\uparrow \hat{z}$



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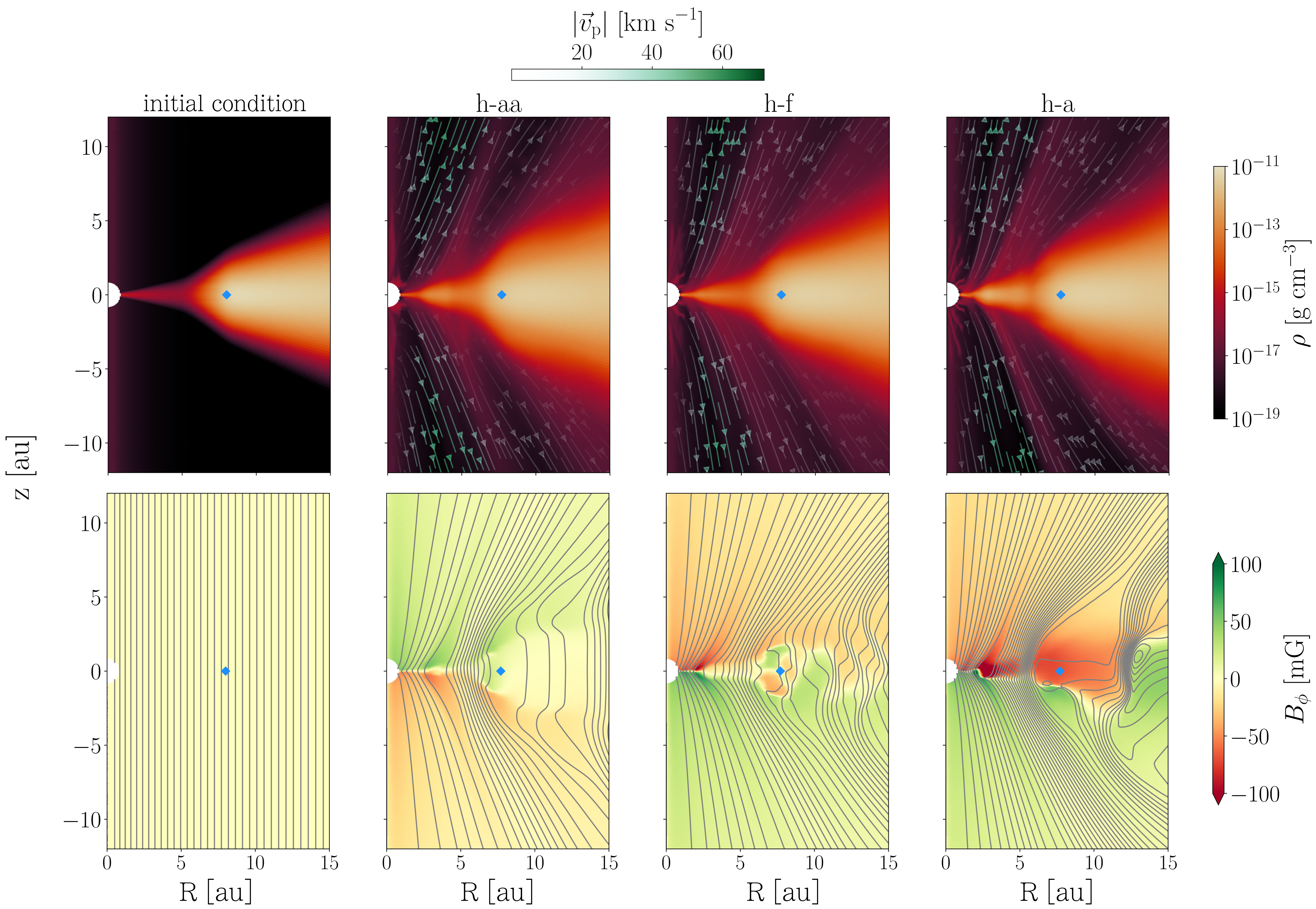

- Same initial condition, different HE configuration
- Cavity and disk survive for all runs, with winds in both
- h-aa → weakest Bφ
  h-a → strongest

$$\frac{\partial B_\phi}{\partial t} \simeq B_R R \frac{\partial \Omega}{\partial R}$$

results

accretion rates - radial variability

h-aa : Hall anti-aligned, $\mathbf{B} \downarrow\uparrow \hat{z}$

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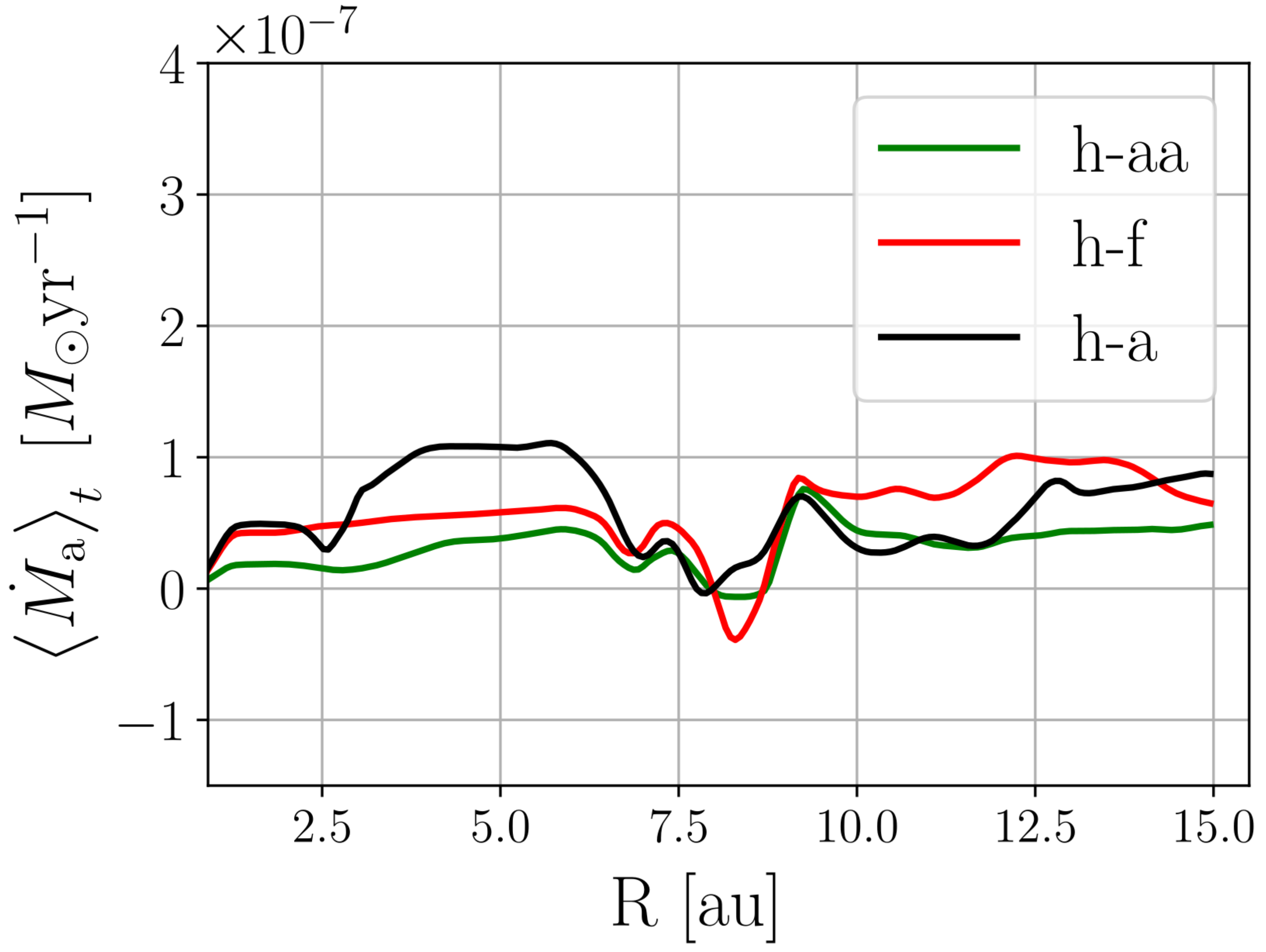

- $\dot{M}_a$ comparable in cavity and disk

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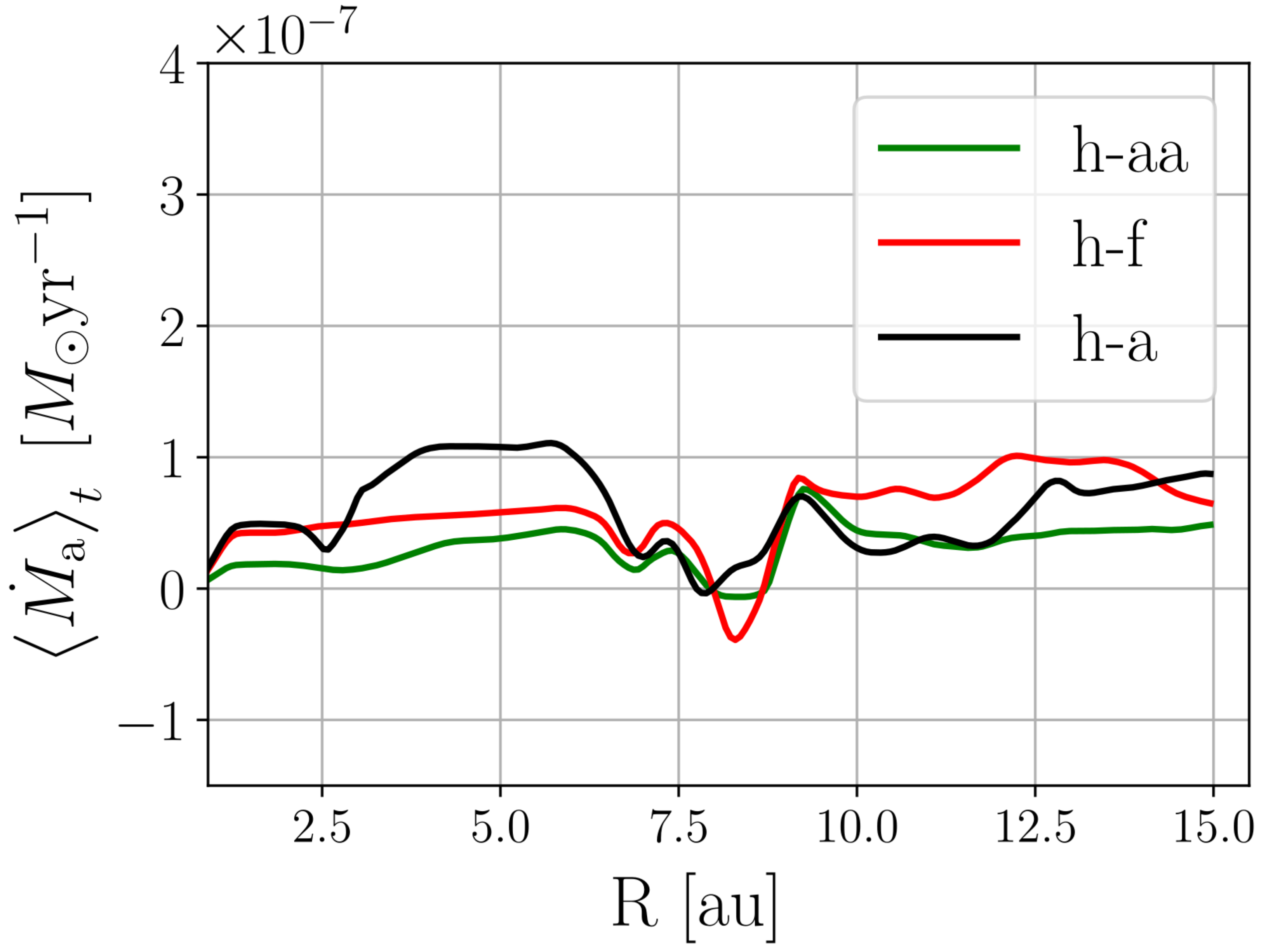

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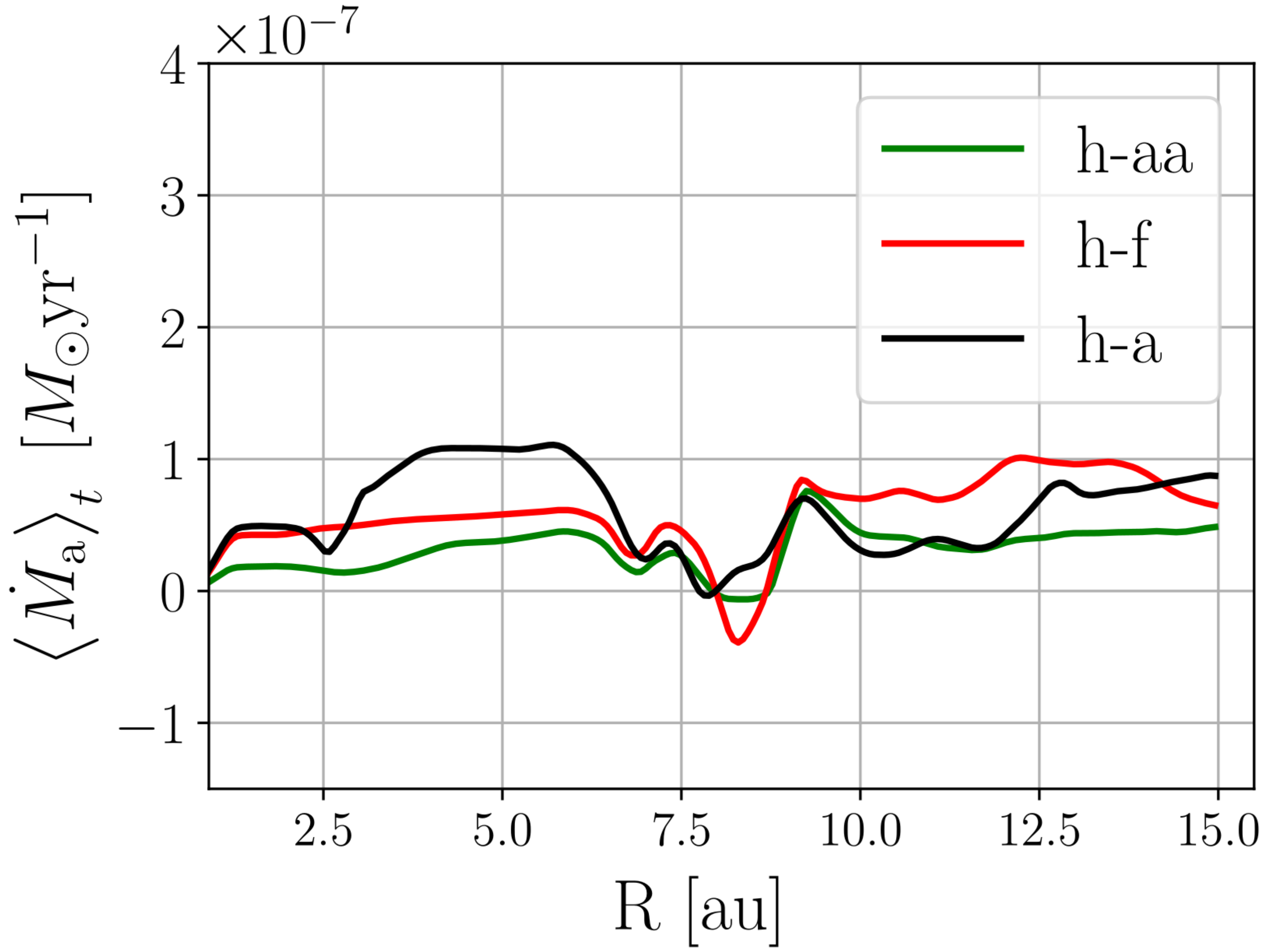

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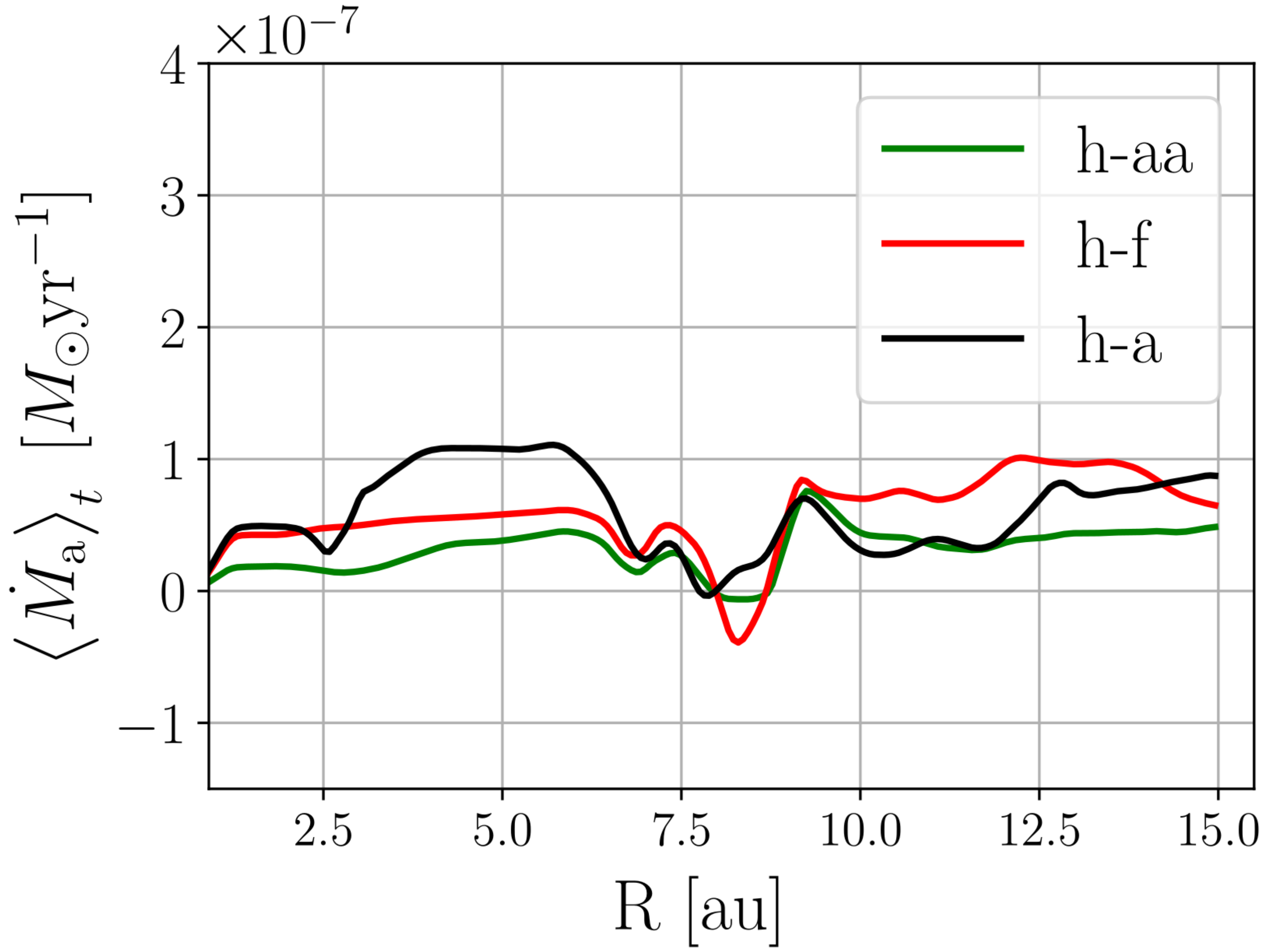

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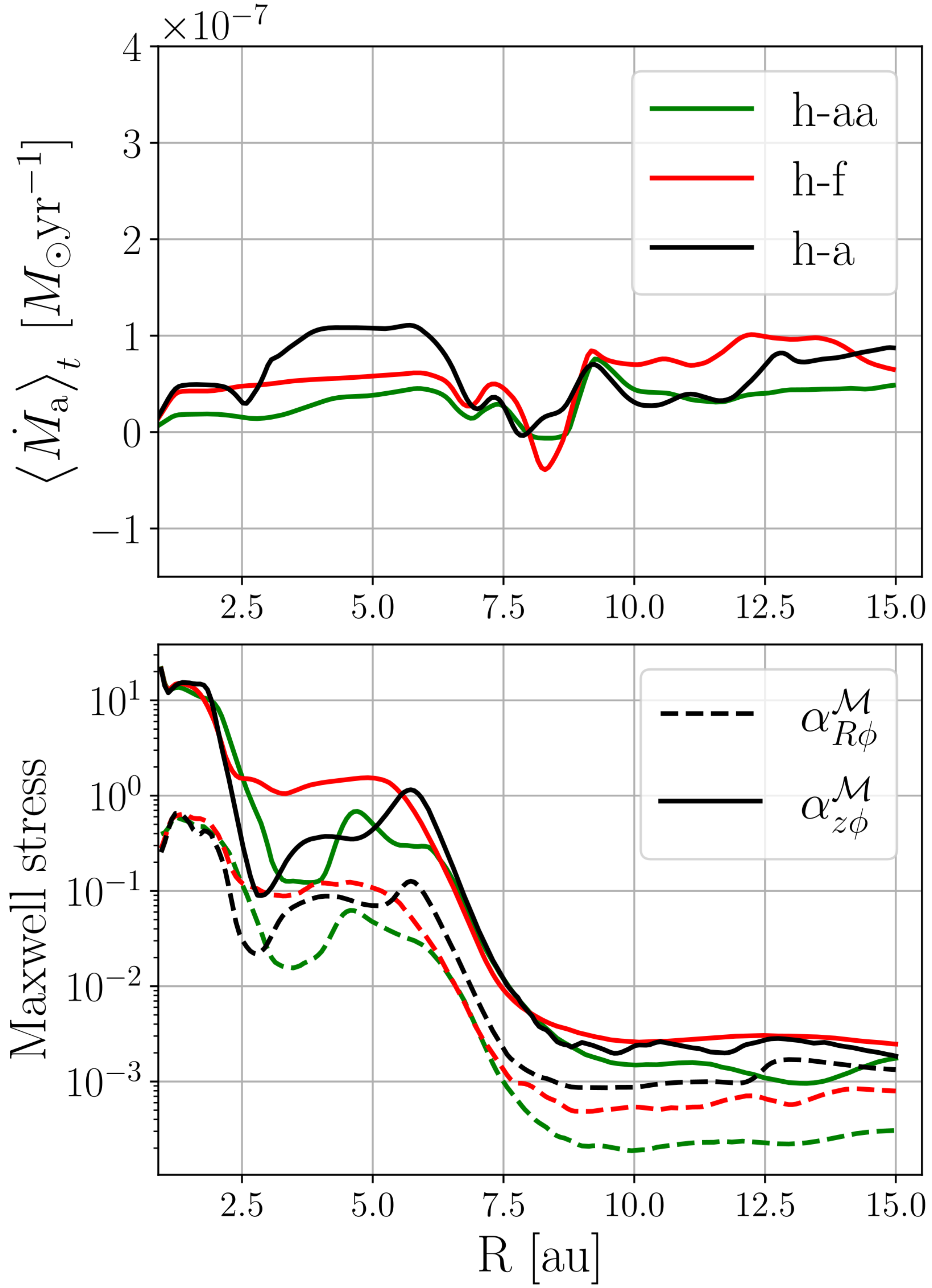

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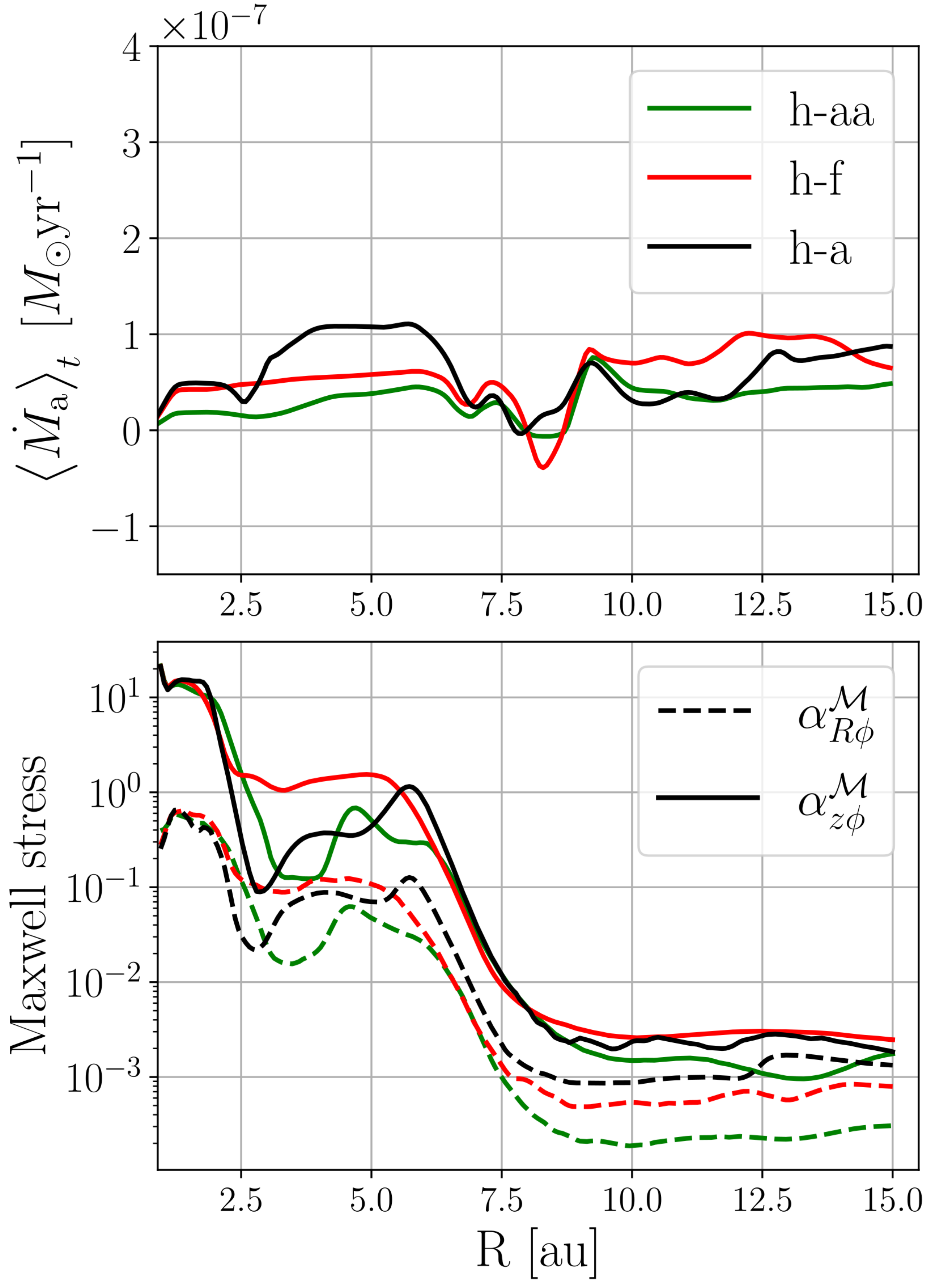

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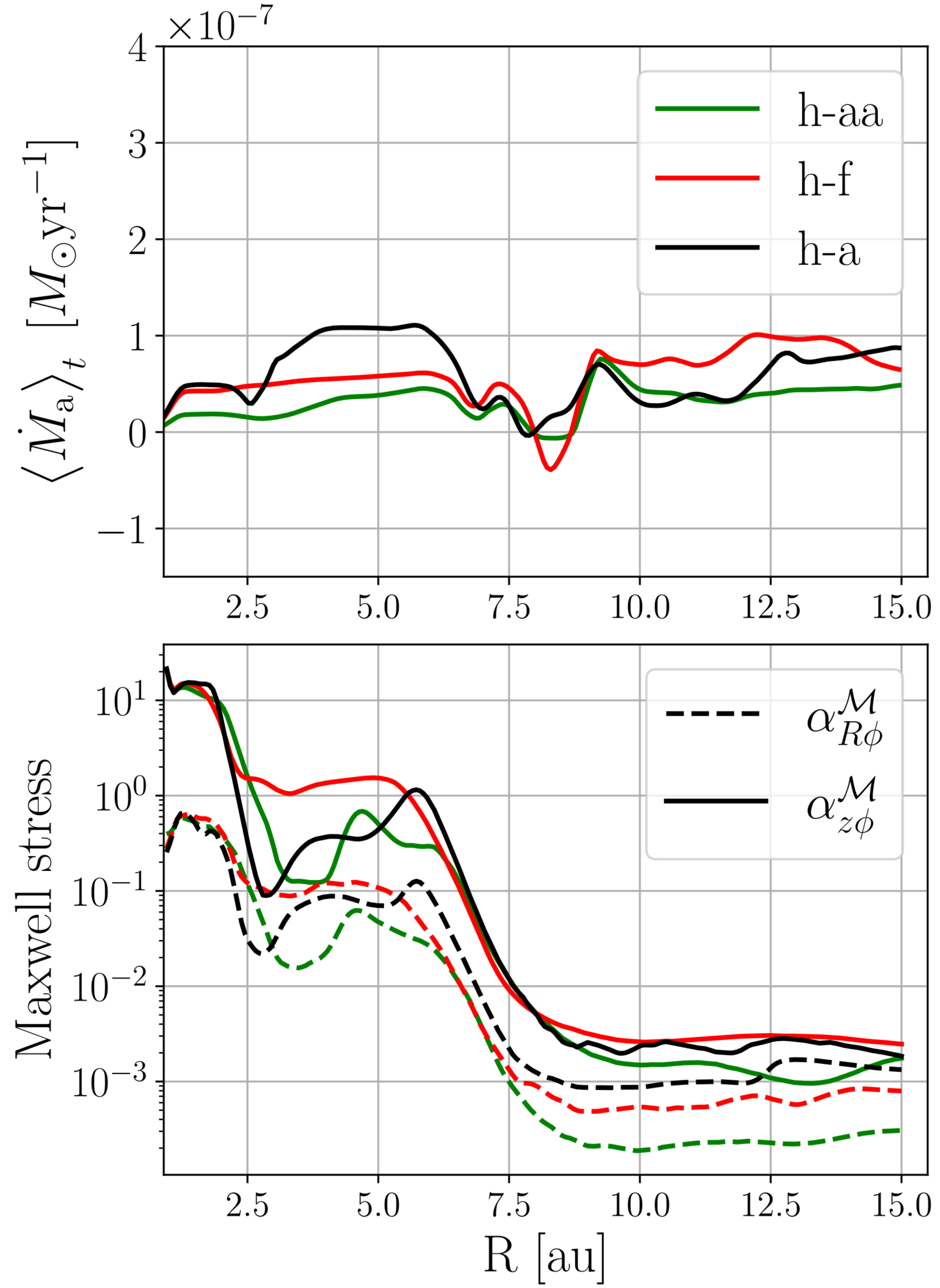

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- overall $\dot{M}_a = (2-8) \times 10^{-8} M_\odot/yr$, comparable with “full disks” - in agreement with works that claim so *(eg. Manara et al. 2014; Owen 2016)*.

results

(near) transonic accretion

- $\dot{M}_a = 2\pi R\overline{\rho v_R}$: If $\dot{M}_a \sim 10^{-8} M_{\odot}/yr$ but $\Sigma_{td}/\Sigma_{fd} \sim 10^{-2}$ then $v_R \sim c_s$ *(Wang & Goodman 2017)*.

  Importance:
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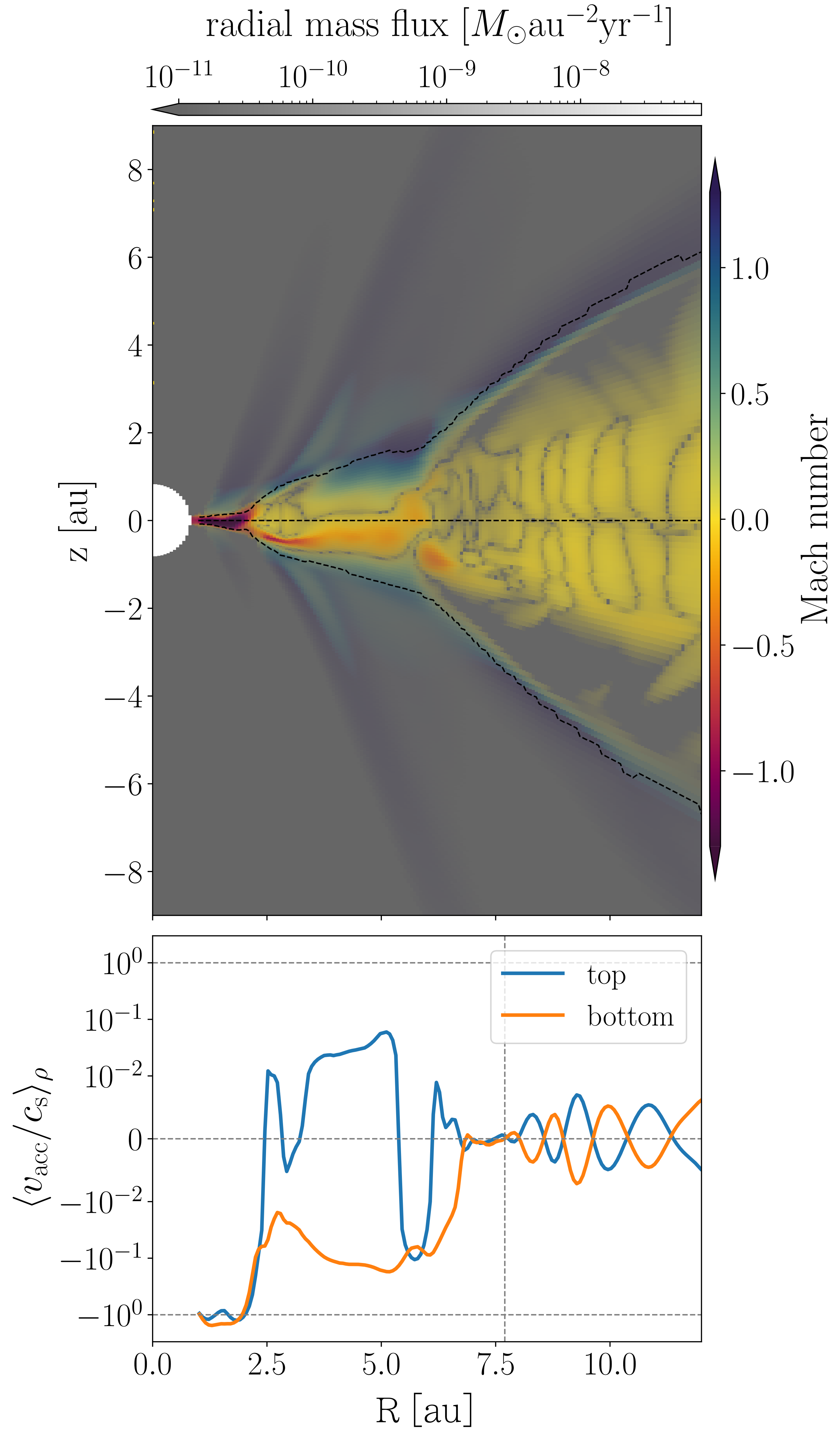

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- Accretion happens away from the midplane, in a preferred region of the disk

- Accretion velocity profile $\rightarrow$ clear transition from subsonic flow in the disk to near-sonic flow in the central cavity.

- For R < 2.5 au we observe transonic accretion - we recover in our simulations a theoretical expectation

results

structure formation

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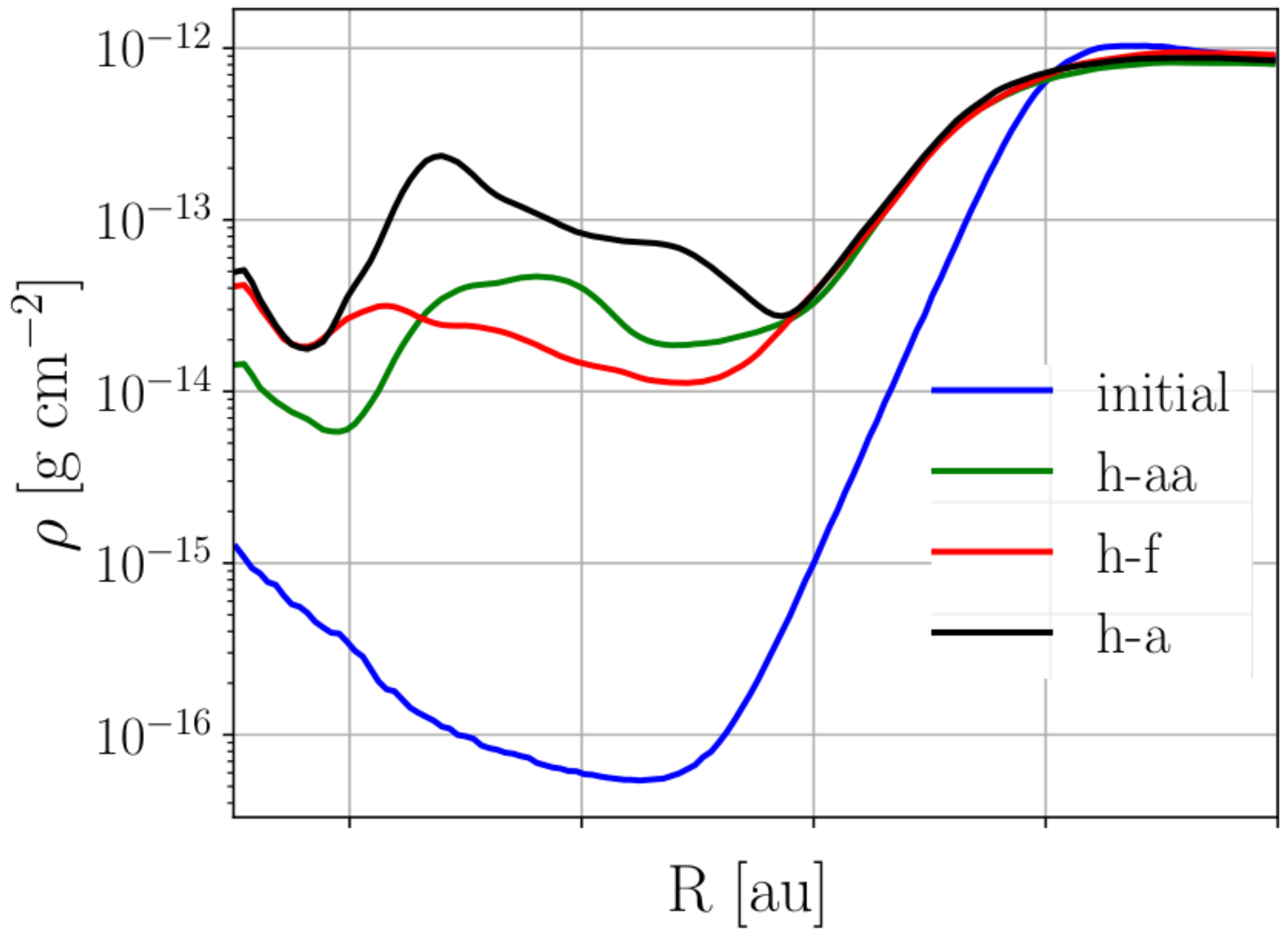

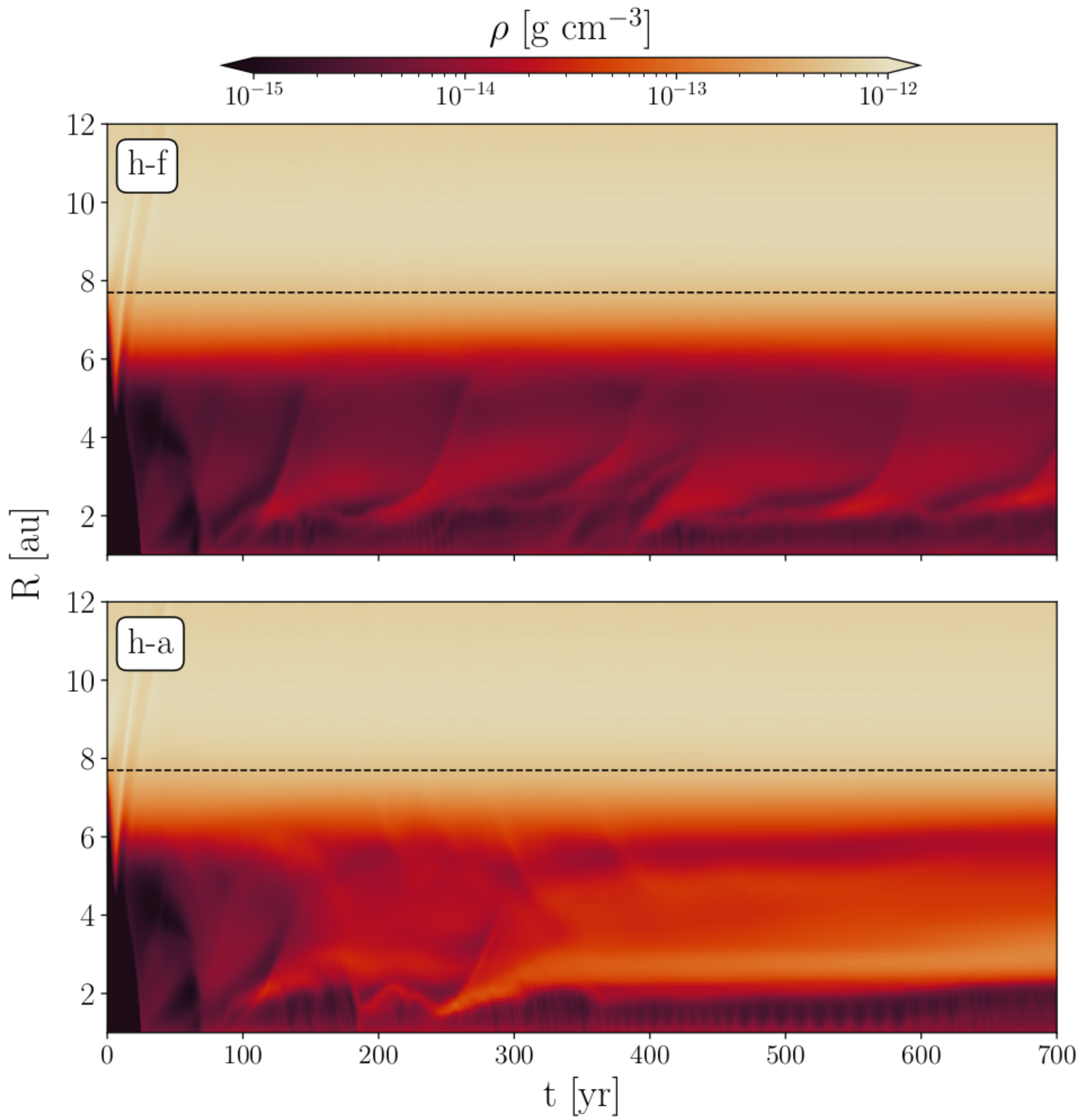

- The cavity filled - but still falls within observational estimates of $\leq 10^{-2}$ *(van der Marel et al 2018)*
- Over-density in the Hall cases

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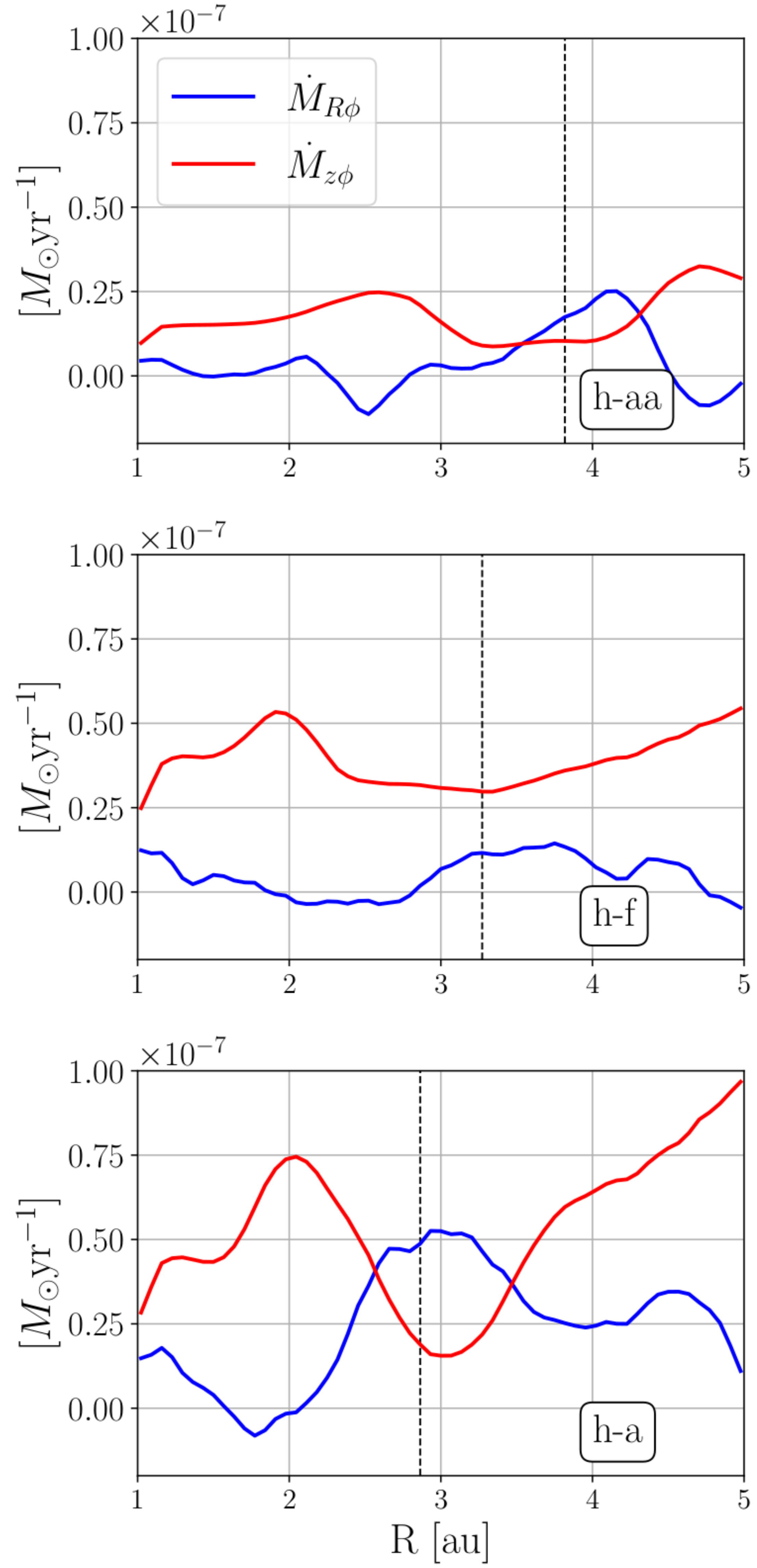

In regions with overdensity, only in the Hall cases $\dot{M}_{radial} > \dot{M}_{wind} \rightarrow$ more mass is replenished in the gap than is being ejected by the wind.

“inverse” **Wind driven instability** *(Suriano et al. 2017; Riols & Lesur 2019)*:

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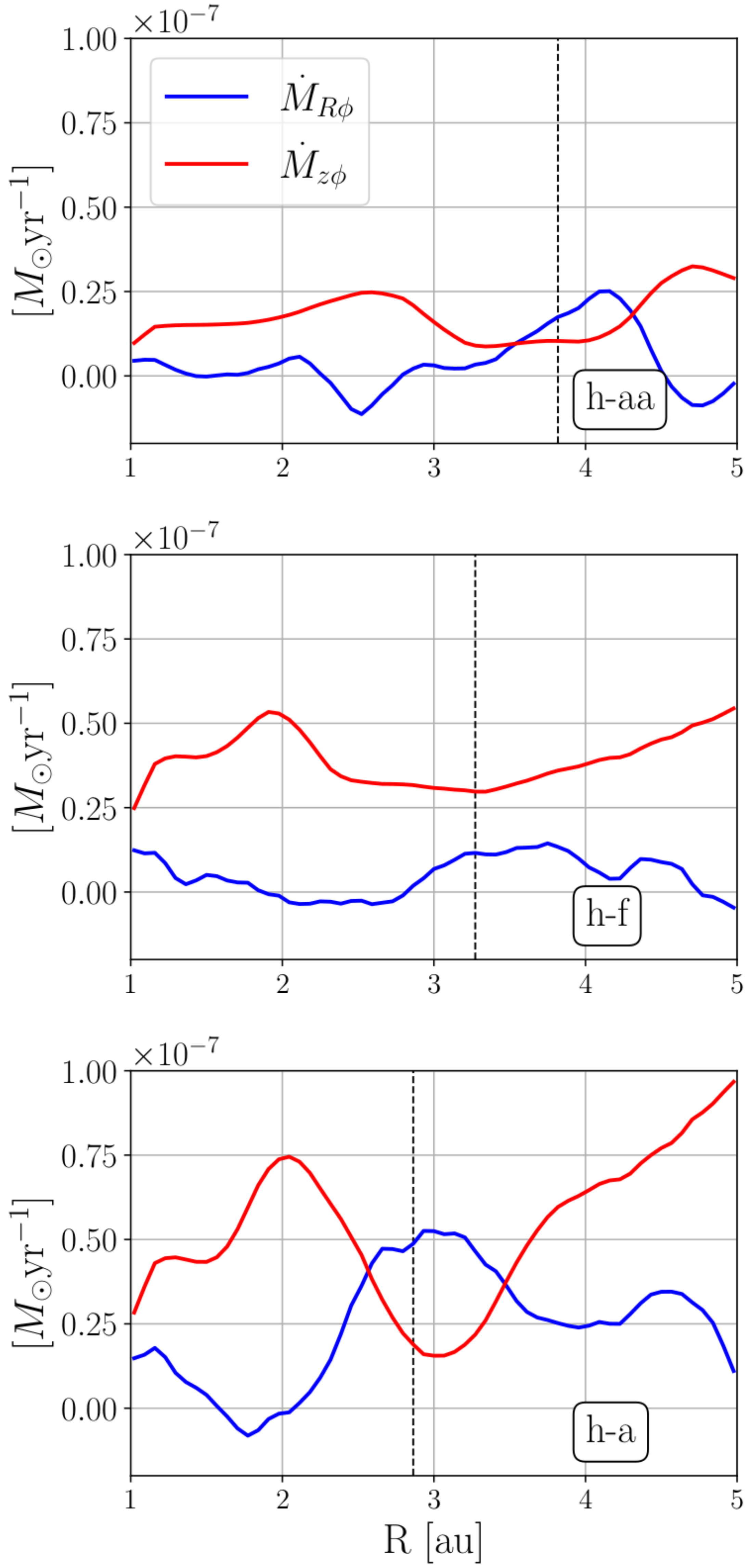

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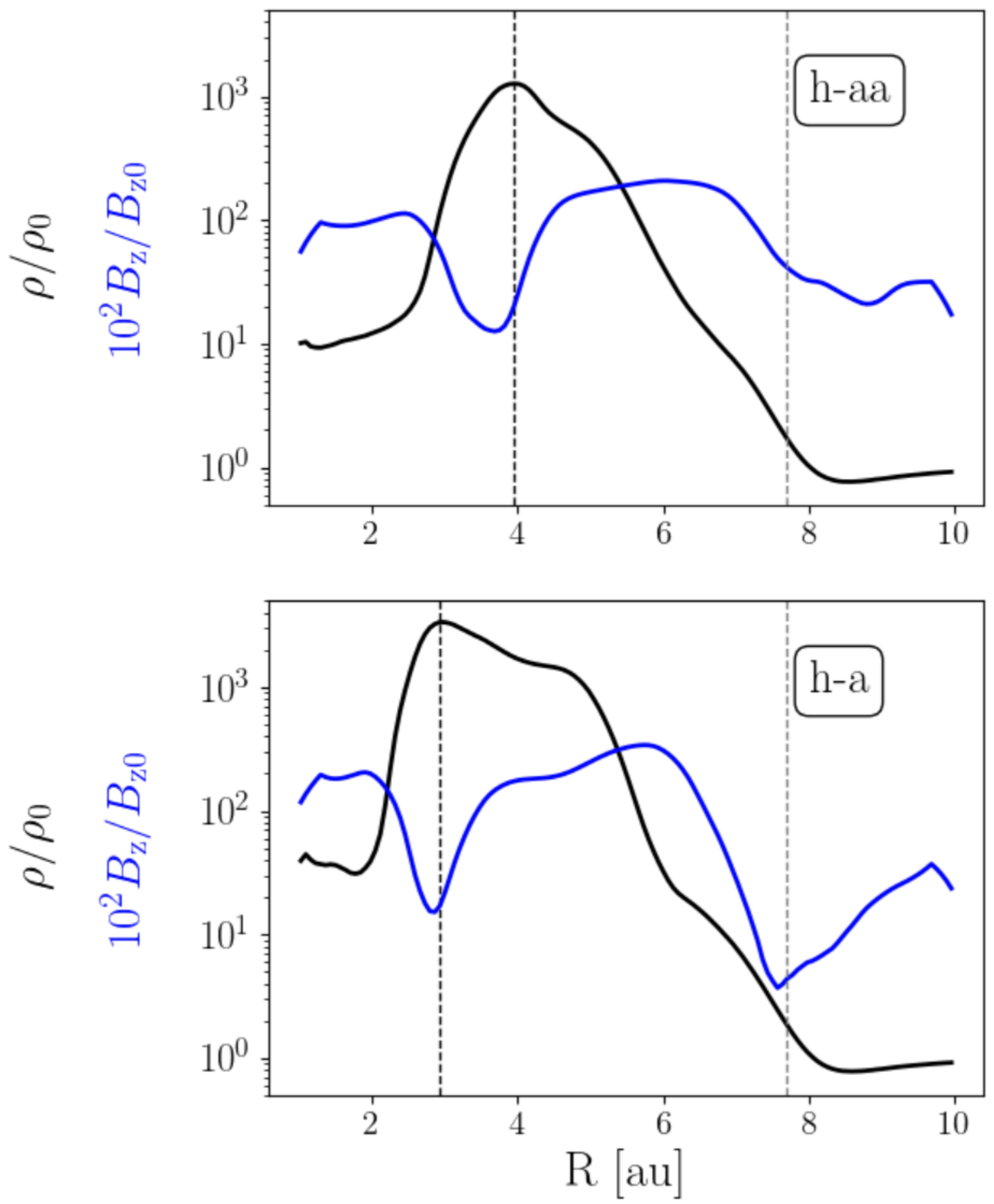

Anti-corellation of ρ, B_z, seen in various works eg. *Kunz & Lesur 2013, Bai 2014, Bethune et al 2017, Riols & Lesur 2019* and more

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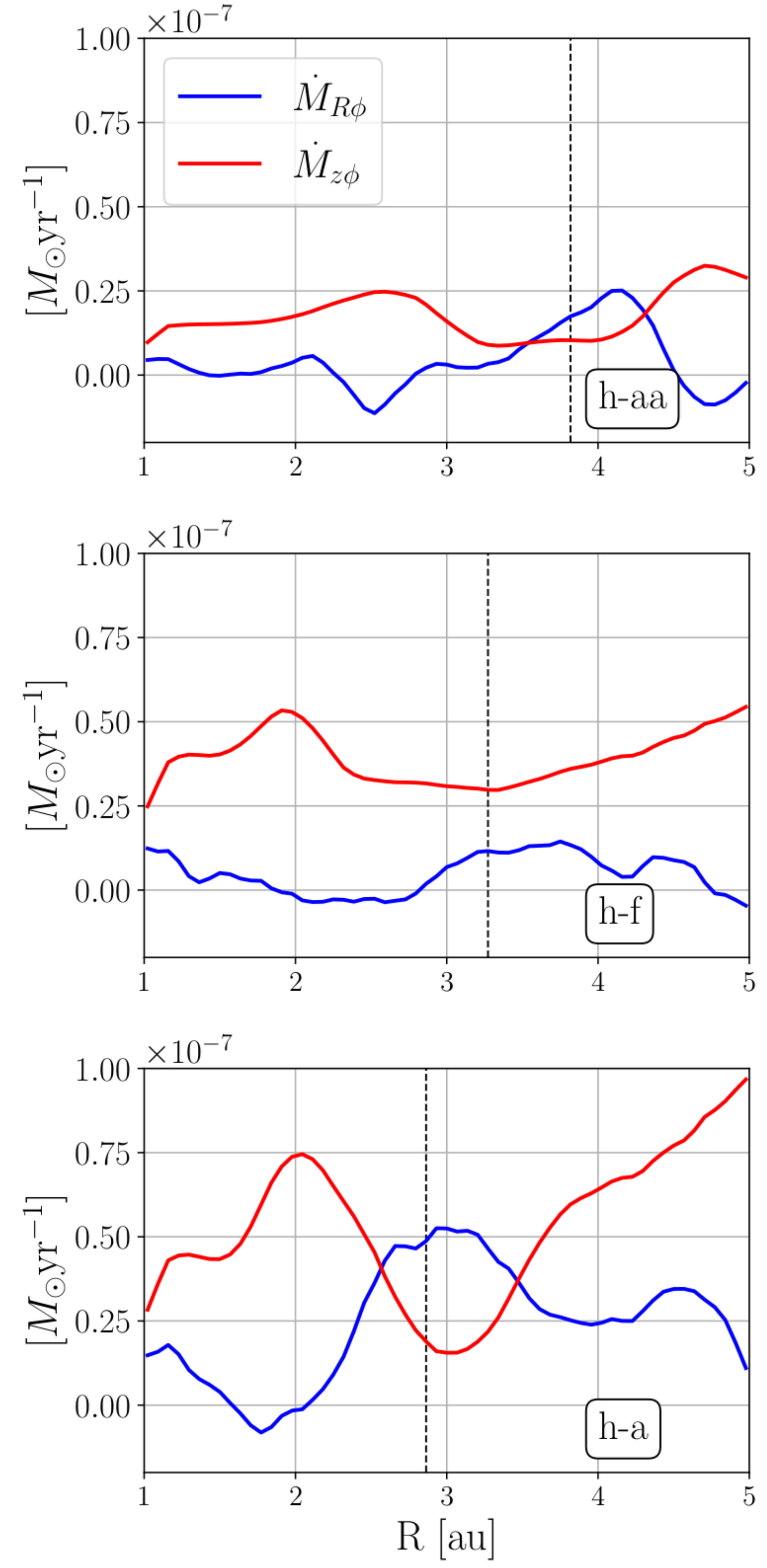

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"inverse" **Wind driven instability** *(Suriano et al. 2017; Riols & Lesur 2019)*:

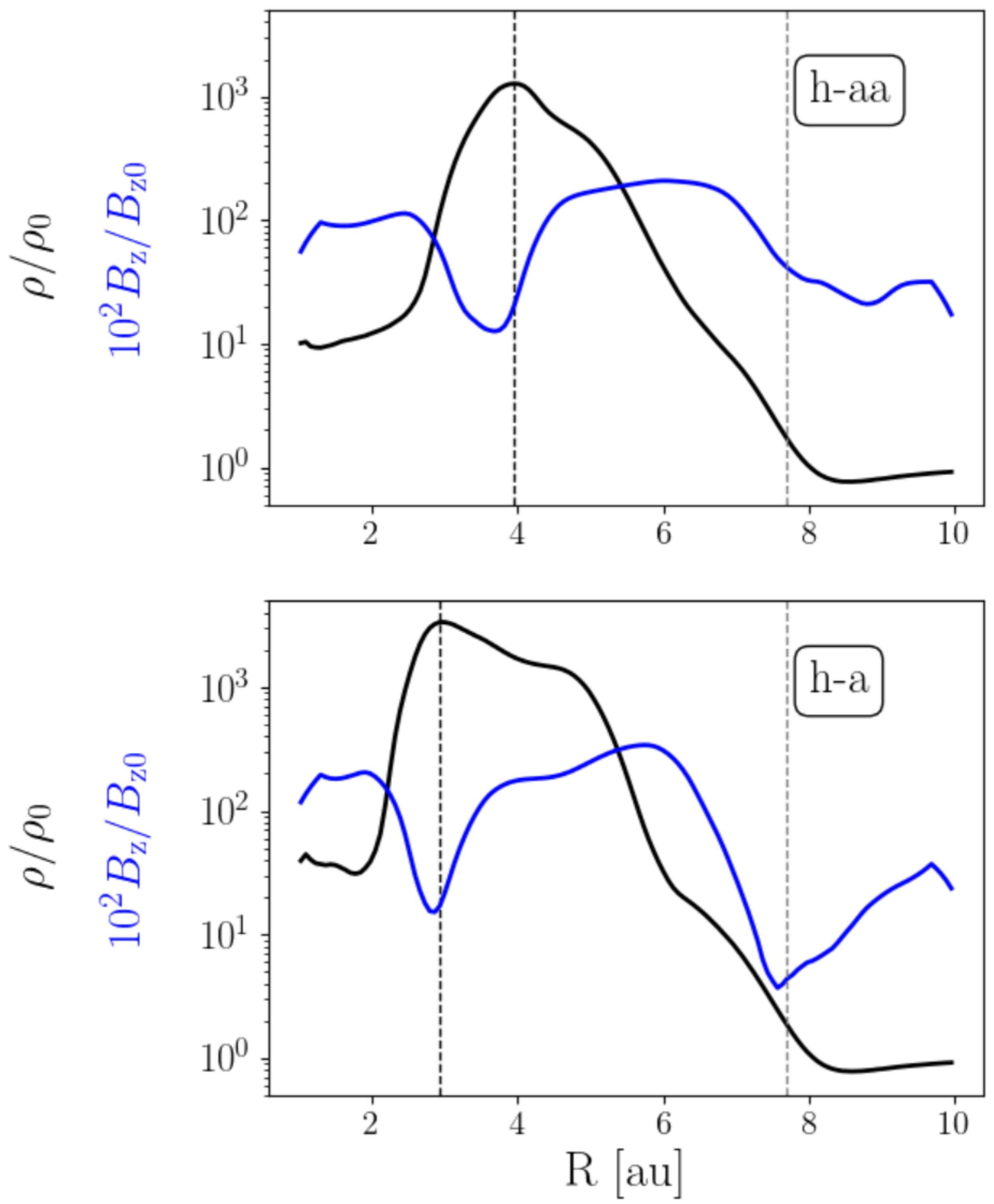
$\dot{M}_{radial} > \dot{M}_{wind} \rightarrow$

mass pile up. $\rightarrow$

magnetic flux evacuation$\rightarrow$

$\dot{M}_{wind} \nearrow \quad \rightarrow$

more prominent structure

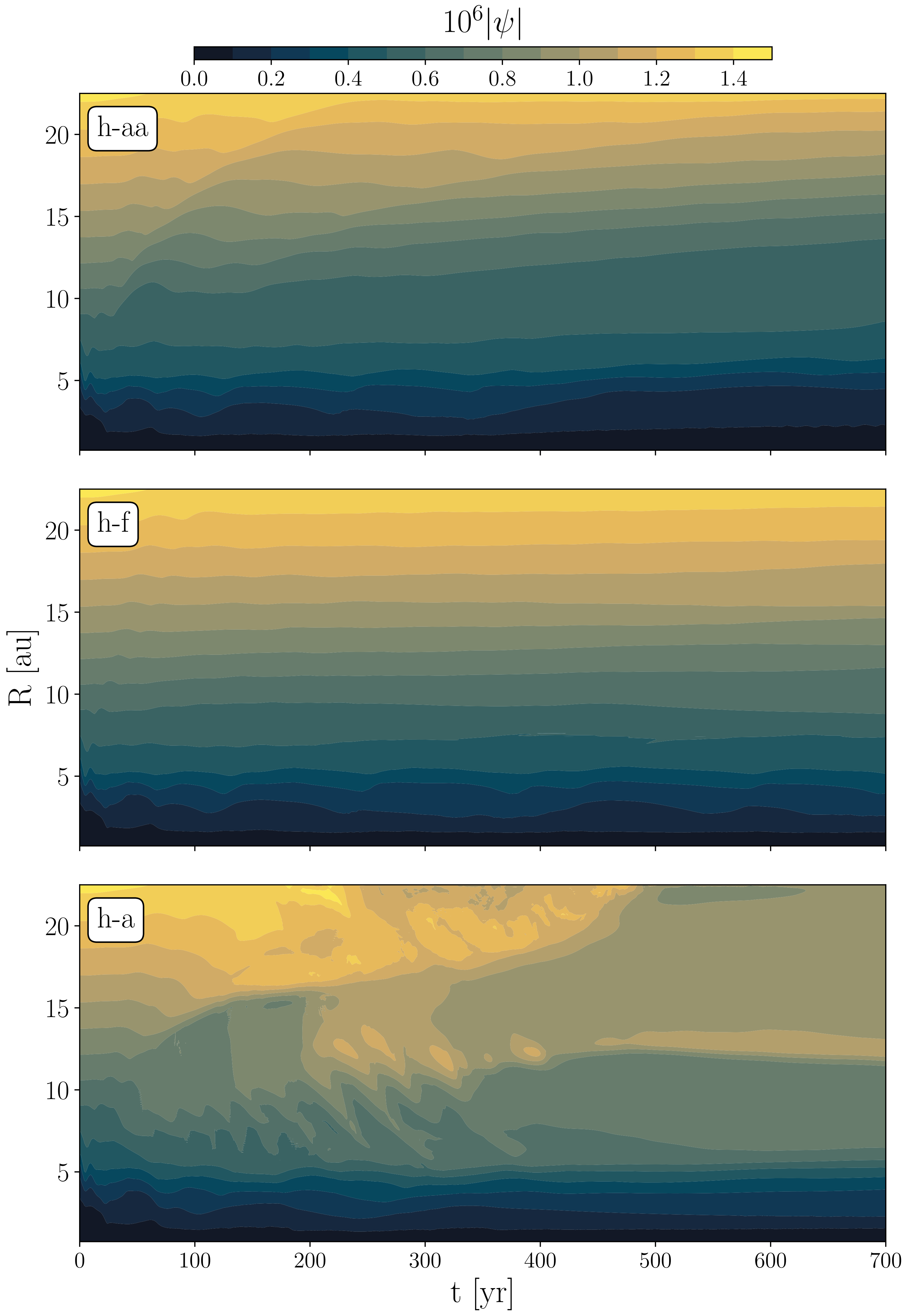


Anti-corellation of ρ, B_z, seen in various works eg. *Kunz & Lesur 2013, Bai 2014, Bethune et al 2017, Riols & Lesur 2019* and more

results

magnetic topology and flux evolution

h-aa : Hall anti-aligned, $\mathbf{B} \downarrow\uparrow \hat{z}$
h-f : only OR and AD, Hall free
h-a : Hall aligned, $\mathbf{B} \uparrow\uparrow \hat{z}$



- accretion/ disk dispersal $\propto$ magnetic field $\rightarrow$ magnetic flux is important

- $\psi = r\, sin\theta\, A_\phi$, the poloidal magnetic flux function - isocontours of ψ describe poloidal field lines

- h-aa, h-f : slow outwards magnetic flux transport - faster in the outer disk than the cavity - v_B quite lower compared to *Bai & Stone 2017*, *Gressel et. al 2020* completely opposite to *Martel & Lesur 2022* who find inward transport

- h-a: similar behaviour of cavity - sudden expulsion of magnetic flux in outer disk - unknown why

final remarks

conclusions

- In all cases: sustained cavity, outer disk behaves like a typical MHD wind driven disk
- Increased accretion rates from h-aa → h-f → h-a
- Transonic accretion — implications for accretion mechanism
- Density bumps within the cavity — potential dust trap

Caveats/ open questions

- Can HE create pressure bumps in any gap setting?
- Magnetic flux transport in the disk - everyone disagrees